

Solving Exponential Equations with Common Bases

Answer Key

Precalculus

Quick Answers

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|----------------------------------|--|
| 1. 3 | 6. (a) $P(5) = 1600$, (b) $t = 6$ |
| 2. 3 | 7. -2 |
| 3. (a) $f(5) = 32$, (b) $x = 8$ | 8. $-1, 4$ |
| 4. 5 | 9. (a) $h(-3) = \frac{1}{27}$, (b) $x = -4$ |
| 5. $\frac{5}{3}$ | 10. 2 |
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Curriculum

Level: Precalculus

HSF-IF.A – problems 3, 6, 9

HSF-LE.A.4 – problems 1, 2, 4, 5, 7, 8, 10

Difficulty 1–5 of 5 across 13 tagged checks.

1. Solve for x : $3^{x+2} = 3^5$

Solution: Both sides are already powers of 3. Because $b^M = b^N$ with $b > 0$, $b \neq 1$ forces $M = N$, set the exponents equal: $x + 2 = 5$, so

$x = 3$

2. Solve for x : $4^x = 64$

Solution: Rewrite the right side with base 4: $64 = 4^3$. The equation becomes $4^x = 4^3$, so $x = 3$.

$x = 3$

3. The table gives values of $f(x) = 2^x$ for $x = 1, 2, 3, 4$. (a) Find $f(5)$. (b) Write 256 as a power of 2 and solve $2^x = 256$.

Solution (a): Each row multiplies the previous value by 2, so $f(5) = 16 \cdot 2 = 32$. Directly, $f(5) = 2^5 = 32$.

$f(5) = 32$

Solution (b): Keep doubling past the table: $2^5 = 32$, $2^6 = 64$, $2^7 = 128$, $2^8 = 256$. So $2^x = 2^8$ and the exponents match.

$x = 8$

4. Solve for x : $5^{2x-1} = 5^{x+4}$

Solution: Same base on both sides, so equate exponents:

$$2x - 1 = x + 4$$

$$x = 5$$

Check: $5^{2(5)-1} = 5^9$ and $5^{5+4} = 5^9$.

$$x = 5$$

5. Solve for x : $8^x = 32$

Solution: Neither side is a power of the other, but both are powers of 2: $8 = 2^3$ and $32 = 2^5$.

$$(2^3)^x = 2^5$$

$$2^{3x} = 2^5$$

$$3x = 5$$

So

$$x = \frac{5}{3}$$

6. $P(t) = 50 \cdot 2^t$ cells after t hours. (a) Find $P(5)$. (b) Solve $50 \cdot 2^t = 3200$.

Solution (a): $P(5) = 50 \cdot 2^5 = 50 \cdot 32 = 1600$ cells.

$$P(5) = 1600$$

Solution (b): Divide by the initial amount first, so the equation becomes a pure power of 2: $2^t = \frac{3200}{50} = 64 = 2^6$, hence $t = 6$ hours.

$$t = 6$$

7. Solve for x : $9^{x-1} = 27^x$

Solution: Both 9 and 27 are powers of 3: $9 = 3^2$, $27 = 3^3$.

$$3^{2(x-1)} = 3^{3x}$$

$$2x - 2 = 3x$$

$$-2 = x$$

So

$$x = -2$$

8. Solve for x : $7^{x^2} = 7^{3x+4}$

Solution: The bases already match, so the exponents are equal — and this time the resulting equation is quadratic.

$$x^2 = 3x + 4$$

$$x^2 - 3x - 4 = 0$$

$$(x - 4)(x + 1) = 0$$

Both roots are legitimate: an exponent may be any real number, so neither is extraneous.

$$x = -1 \quad \text{or} \quad x = 4$$

9. The table gives values of $h(x) = 3^x$. (a) Find $h(-3)$. (b) Solve $3^x = \frac{1}{81}$.

Solution (a): Moving left in the table divides by 3 each step, and a negative exponent is a reciprocal: $h(-3) = 3^{-3} = \frac{1}{3^3} = \frac{1}{27}$.

$$h(-3) = \frac{1}{27}$$

Solution (b): Write the right side as a power of 3: $81 = 3^4$, so $\frac{1}{81} = 3^{-4}$. Then $3^x = 3^{-4}$ and the exponents match. $x = -4$

10. Solve for x : $\left(\frac{1}{2}\right)^{x+1} = 8^{x-3}$

Solution: A reciprocal base is a negative exponent: $\frac{1}{2} = 2^{-1}$, and $8 = 2^3$.

$$(2^{-1})^{x+1} = (2^3)^{x-3}$$

$$2^{-x-1} = 2^{3x-9}$$

$$-x - 1 = 3x - 9$$

$$8 = 4x$$

So

$x = 2$